Predicting shear stress based on labquakes

Earthquake physics and ML's project

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1 Abstract

Shear stress. *The shear stress is the stress component complanar to the cross section.*

Our idea is to create a model that, given the acoustic emissions of a laboratory earthquakes (as the ones from the Biax esperiment at Penn State University), tries to infer the shear stress that the fault zone is undergoing (similar to what is done in [5]).

As stated in [5], acoustic emissions seems to be a fingerprint of fault zone stress: maybe, this kind of approach, using online data, can be used to do real time evaluations of the fault zone stress?

Further on, it would be interesting to try and apply our model to real earthquakes data: past works, like [1], highlight possible breakthroughts in this direction, as real world faults and earthquakes do produce data that can be considered analogous to labquakes's acoustic emissions.

2 Introduction

Most tectonic earthquakes take place when juxtaposed crustal blocks overcome the static fault friction and abruptly slide past one another, releasing an immense amount of energy in the form of seismic waves. Therefore, a rupture initiates and propagates along the fault plane. It is the frictional state that controls how the fault ruptures, its nucleation, how big the earthquake will become and even when the next event may take place.

As stated in [5], recent works show a fundamental relation between the sound emitted by the fault and its physical state, which is strictly related to the fault zone shear stress and tells us how close it is to failing.

Our goal is to show that the most important physical properties of a fault can be accurately estimated using ML to analyze the sound that the fault broadcasts in a laboratory setting that closely mimics Earth faulting.

3 Model

Our purpose with the model is to infer the shear stress (the stress component complanar to the cross section) that the fault is undergoing. The shear stress is related to the labquake (and earthquake) cycle, as it reaches its maximum right before a seismic event occurs: understanding how the shear stress evolves can give us a better understanding on how the cycle works, and help us in predicting new events.

It is important to note that we will only be focusing on prediction (and not on forecast): we will be using data up to time t in order to evaluate the shear stress at time t. While having the same input, the forecasting problem has different outputs: it aims to predict the shear stress starting from time t.

To infer the stress we used acoustic emission retrieved from the experimental setting Biax at Penn State University (experiments p4581, p4679 and p5198 from [6]) and some models inspired by the one proposed in [5]. We tried to make some minor adjustments to the model architecture, to see if we could improve in any way its performances.

The main components of this model are:

- **LSTM** (Long Short-Term Memory) layer, a recurrent layer that takes in input a sequence of elements and update its internal state (as described in [3];
- 3 **convolutional** layers;
- 3 **dense** layers, that regress to shear stress itself.

As per the experimental setting, the input data consists of acoustic emissions stored in two different features (named `Var_Ch1` and `Var_Ch2`), as they come from two different piezoelectric sensors. The model in 1 takes in input both channel together (by using a single **Input** channel, and a single **LSTM** layer). This architecture could lead to "contamination" between the channels: we tried to assess whether this contamination exists, and whether it is useful or detrimental to the accuracy of the model, by training a double input channel model.

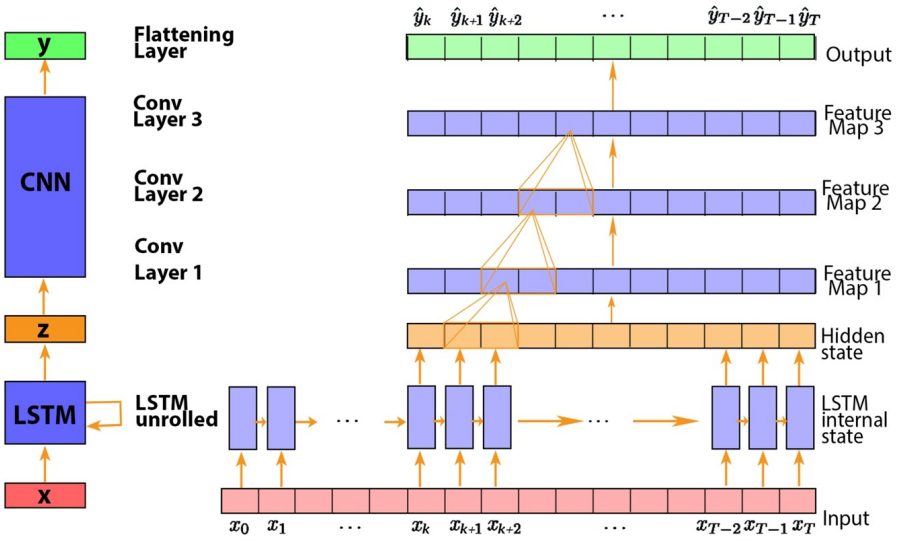


Figure 1: the architecture from Laurenti et al.

To double input channels, two different **LSTM** layers are required: their outputs (as in their internal recurrent states) are then concatenated and given as an input to the next layer.

So, we trained 3 different models:

- a single channel and experiment specific one (heavily inspired by the original one) with an additional logarithmic layer (*log* normalization) to normalize the input;
- a double channel, experiment specific one, without the *log* normalization (as it significantly slowed the training, in terms of convergence speed);
- the final model, experiment agnostic and double channel.

All the models share the same input and output: the single and double channel is handled by the model itself.

Why an experiment agnostic model? [5] stated that the best value for the `units` hyperparameter is related to the seismic cycle length (that is different based on different experimental settings). This parameter cannot be dinamically implemented, so the model cannot be adjusted to the experiment the data is being picked from. But, given that the underlying physics is the same, is there a way to generalize the model?

We thought of training more specific models, but each specific model had very poor performances on unseen data: maybe, a more general model could have good performances while also being more memory efficient than lots of specific models. We tried to create a training set with data from all the experiments we could access (that was still of manageable size) and got some interesting results.

All model are built using the library *Keras* [4].

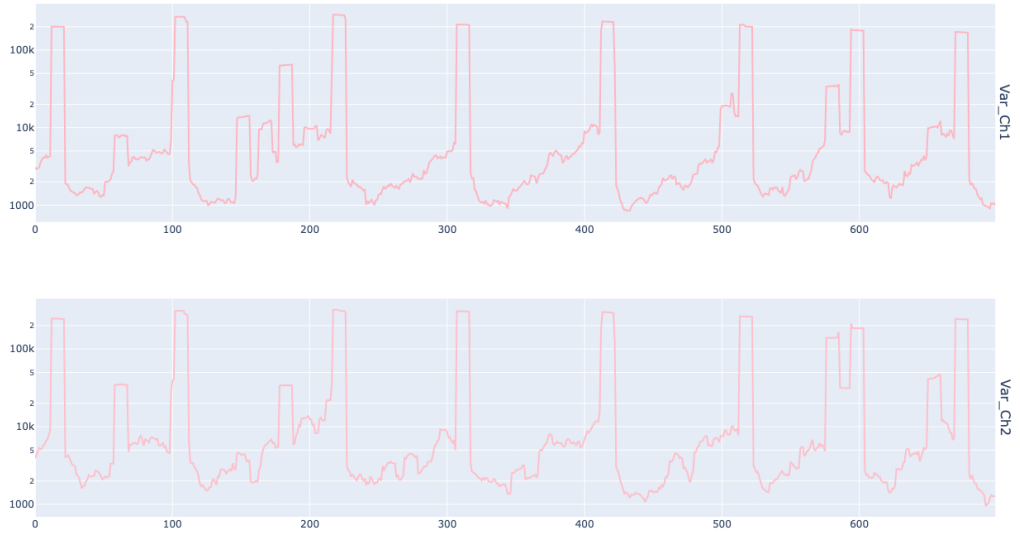


Figure 2: acoustic emissions from p4581, samples from 0 to 700

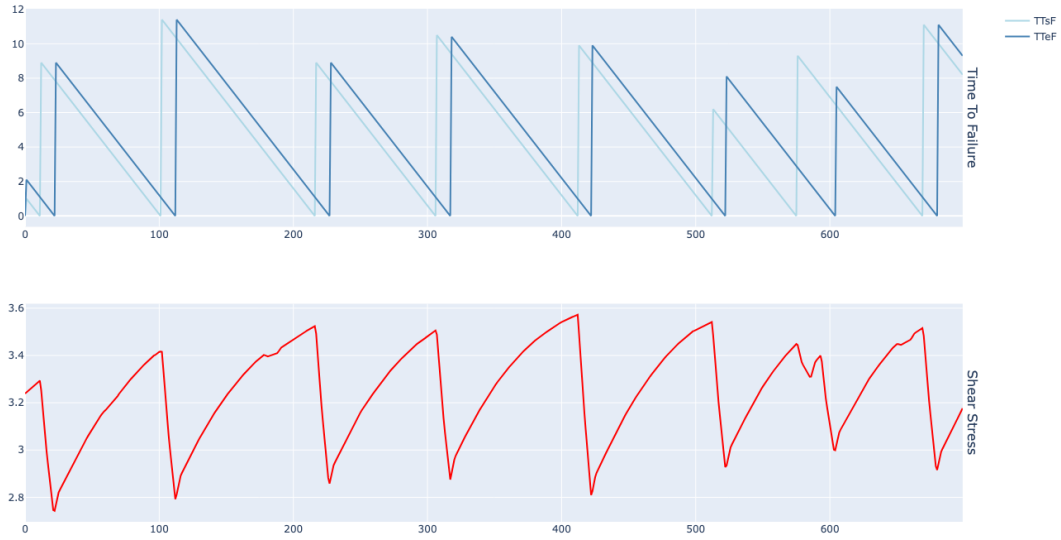


Figure 3: shear stress from p4581, samples from 0 to 700

4 Results

4.1 Single channel, experiment specific model

This first model is composed of the **Input** layer, that applies the logarithm to the input: we found out that this additional transformation speeds up the convergence. Then, there is the **LSTM** layer, 3 **convolutional** layers and 3 more **dense** (fully connected) layers: the final output has only one dimension.

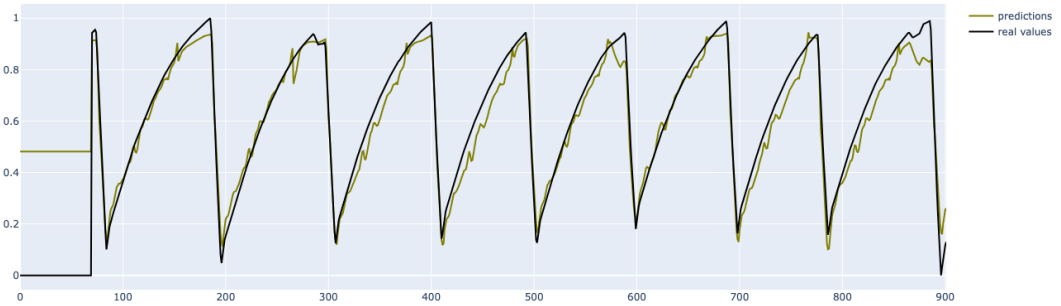


Figure 4: predictions from the first model on p4581

Our version of this model has a RMSE = 0.1433 on the test set (from the same experiment as the training set).

4.2 Double channel, experiment specific model

Here, the input is divided in two channel, that then go through two different **LSTM** layers: their output is then concatenated and given to 2 **convolutional** layers and 3 **dense** ones. We found out that there was need neither for the third convolutional layer, nor the logarithmic normalization.

On the same test set as before, we get a RMSE = 0.0629 and a slightly better performance, especially on the uprising fraction of the seismic cycle.

But what about new and unseen data?

Plots from figure 6 show that both models do not generalize well to unseen data; they also show that the model is highly dependent on the experimental setting and the materials used (as p4581 and p5198 have similar experimental settings, and p5198 performs significantly better than p4679).

4.3 Double channel agnostic model

The performances of the double channel model were promising: this last model has the same architecture, but it is being trained on a new training set (that includes data points

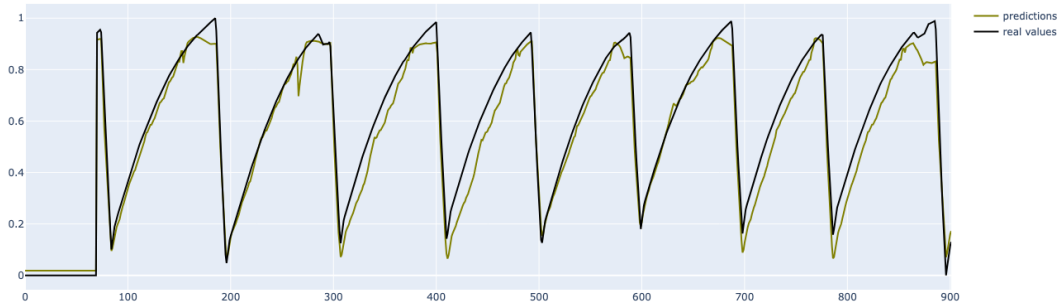


Figure 5: predictions from the second model on p4581

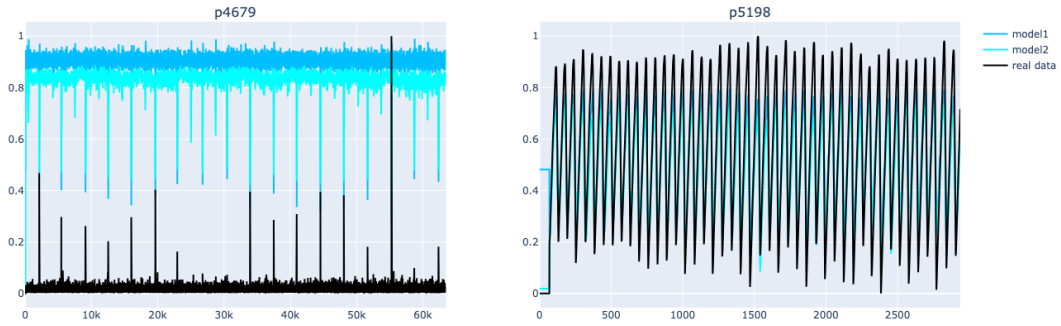


Figure 6: predictions on p4679 and p5198

	single channel	double channel
p4679	0.8879	0.8193
p5198	0.1292	0.1365

Table 1: RMSE over different experiments and models

from all previous experiments). Maybe it will help the model better generalize?

And what about the RMSE?

While the result are promising, we must admit that the double channel model is sligthly bigger than the single channel one (as it has more weights). We suggest that there are still some improvements to it, but we cannot demonstrate that those improvements do not only come from the model being bigger.

All code is available on GitHub.

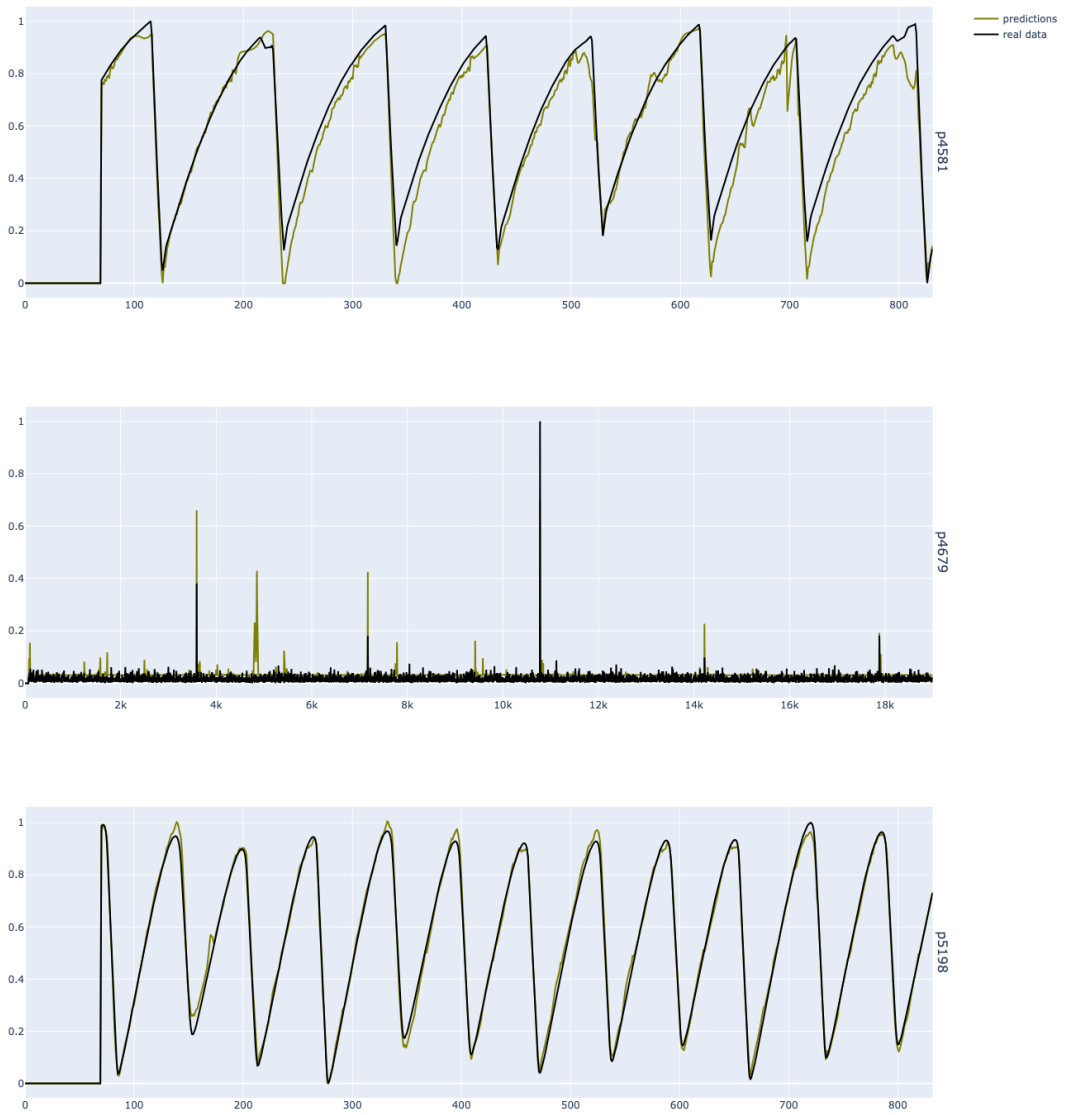


Figure 7: predictions on p4581, p4679 and p5198 from the third model

5 Working with real data

Solid Earth events such as earthquakes, volcanoes and tsunamis can generate atmospheric and ionospheric perturbations by various coupling (interaction) mechanisms.

During earthquakes, an example of dynamic coupling occurs, thanks to the vertical oscillations of the Earth's surface launch pressure waves in the neutral atmosphere that

Experiment	RMSE
p4581	0.0659
p4679	0.0248
p5198	0.0241

Table 2: RMSE over the experiments using the final model

gain amplitude as they attain ionospheric heights.

In particular, surface Rayleigh waves emanating from an earthquake cause ground displacements that can produce acoustic waves that propagate upwards.

Rayleigh waves are surface waves that propagate through solid materials, they are a combination of both longitudinal and transverse waves so, as they propagate, they create an elliptical motion. These waves can be related to shear stress; in fact, analysing this kind of acoustic signal we can keep track of the shear stress along fault zones.

The approach we thought of implementing was a **domain adaptation** one (like the ones listed in [2]): our idea was to train a model "translating" Rayleigh waves into acoustic emission, and then give this translation in input to the model regressing to shear stress values. We found the challenge to be more difficult that we thought, as there is not enough data available. Still, we think this kind of approach can give interesting results; and we hope to have the chance to try it out.

References

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[2] Abolfazl Farahani et al. *A Brief Review of Domain Adaptation*. 2020. arXiv: 2010.03978 [cs.LG]. URL: https://arxiv.org/abs/2010.03978.

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[6] *Marone Lab Data*. URL: http://psudata.s3-website.us-east-2.amazonaws.com/.